MH3600 Topology and Manifolds

Solutions of Tutorial #7

Topics covered:

- Connectedness

Problem 1

Let $\mathcal{T}$ and $\mathcal{T}'$ be two topologies on X. If $\mathcal{T}' \supset \mathcal{T}$, what does connectedness of X in one topology imply about connectedness in the other?

Solution: Connectedness of X in the finer topology $\mathcal{T}'$ implies connectedness in the coarser topology $\mathcal{T}$. We prove the contrapositive. Suppose there is a separation $A \cup B$ of X in $\mathcal{T}$. Then $A \cup B \in \mathcal{T} \subset \mathcal{T}'$ so X has a separation in $\mathcal{T}'$.

The converse implication does not hold. Indeed, consider the real line $\mathbb{R}$ with both the standard topology and the discrete topologies. Using the standard topology, the real line $\mathbb{R}$ is connected, but it is not connected when using the discrete topology, which is strictly finer than the standard one. ■

Problem 2

Let $\{A_n\}$ be a sequence of connected subspaces of X, such that $A_n \cap A_{n+1} \neq \varnothing$ for all n. Show that $\bigcup A_n$ is connected.

Solution: Assume $\bigcup_n A_n$ has a separation $C \cup D$. Then since each A_n is connected, either $A_n \subset C$ or $A_n \subset D$ for each n. Let k be the smallest element of $\mathbb{Z}_+$ such that $A_k \subset C$ and $A_{k+1} \subset D$. Then $A_k \cap A_{k+1} = \varnothing$, so it must be that there exists no such k. So without loss of generality, $A_n \subset C$ for each n, contradicting the fact that D is nonempty. ■

Problem 3

Let $\{A_\alpha\}$ be a collection of connected subspaces of X; let A be a connected subspace of X. Show that if $A \cap A_\alpha \neq \varnothing$ for all α, then $A \cup (\cup A_\alpha)$ is connected.

Solution: Assume $A \cup (\cup_\alpha A_\alpha)$ has a separation $C \cup D$. Then since each A_α is connected, either $A_\alpha \subset C$ or $A_\alpha \subset D$ for each α. Since A is connected, $A \subset C$ or $A \subset D$. Without loss of generality $A \subset C$. Then $A \cap A_\alpha = \varnothing$ for each α with $A_\alpha \subset D$. So it must be that $A_\alpha \subset C$ for each α, contradicting the fact that D is nonempty. ■

Problem 4

Show that if X is an infinite set, it is connected in the finite complement topology.

Solution: Assume $A \cup B$ is a separation of X. A and B are open, so their complements are finite. Since $B \subset X - A, B$ is finite. Similarly, A is finite. But then X is the union of two finite sets and hence finite, contradicting the fact that X is infinite. $\blacksquare$

Problem 5

Let $A \subset X$. Show that if C is a connected subspace of X that intersects both A and $X - A$, then C intersects $\operatorname{Bd} A$.

Solution: We begin by noting that for all sets $A, \bar{A} = \text{int } A \cup B\,\text{d}A$. Now, assume C does not intersect Bd A. Since C intersects A and $X - A$, C intersects the closures $\bar{A}$ and $\bar{X} - A$. Since $\bar{A} = \text{int } A \cup \text{Bd}A$ and $\overline{X - A} = \text{int } X - A \cup \text{Bd}A$, it follows that C intersects the closures only at the interiors. Since $X = \text{int}A \cup \text{int}X - A \cup \text{Bd}A$, it follows that $C = (C \cap \text{int}A) \cup (C \cap \text{int}X - A)$. Since C intersects both of these sets, neither is empty. But then $(C \cap \text{ int } A) \cup (C \cap \text{ int } X - A)$ is a separation of C, contradicting the connectedness of C. Thus C intersects $\text{Bd}A$. $\blacksquare$

Problem 6

Let A be a proper subset of X, and let B be a proper subset of Y. If X and Y are connected, show that

$$(X \times Y) - (A \times B)$$

is connected. **Hint:** use the conclusion of Problem 3.

Solution: For $x \in X$, let $\ell_x = \{x\} \times Y$, and for $y \in Y$, let $\ell_y = X \times \{y\}$. Notice that both types of sets are connected since they are homeomorphic to either X or Y. Notice further that $(X \times Y) - (A \times B) = \left(\bigcup_{a \in X-A} \ell_a\right) \cup \left(\bigcup_{b \in Y-B} \ell_b\right)$. Fix $x \in X - A$. Then for each $b \in Y - B, \ell_b$ intersects ℓ_x. Then, by Problem 3, $\ell_x \cup \left(\bigcup_{b \in Y-B} \ell_b\right)$ is connected. Denote this set by C. Notice that for all $a \in X - A, \ell_a$ intersects C. Thus, again by Problem 3, $(X \times Y) - (A \times B) = \left(\bigcup_{a \in X-A} \ell_a\right) \cup \left(\bigcup_{b \in Y-B} \ell_b\right) = C \cup \left(\bigcup_{a \in X-A} \ell_a\right)$ is connected. $\blacksquare$

Problem 7

Let $\{X_\alpha\}_{\alpha \in J}$ be an indexed family of connected spaces; let X be the product space

$$X = \prod_{\alpha \in J} X_\alpha$$

Let $\mathbf{a} = (a_\alpha)$ be a fixed point of X.

(a) Given any finite subset K of J, let X_K denote the subspace of X consisting of all points $\mathbf{x} = (x_\alpha)$ such that $x_\alpha = a_\alpha$ for $\alpha \notin K$. Show that X_K is connected.

(b) Show that the union Y of the spaces X_K is connected.

(c) Show that X equals the closure of Y; conclude that X is connected.

Solution: **(a)** Notice that X_K is homeomorphic to $\prod_{k\in K} X_k$ which is connected by Theorem 23.6.

(b) Notice that all of these spaces share the common point $\mathbf{a}$, so that by Theorem 23.3, Y is connected.

(c) Let $\mathbf{x} = (x_\alpha) \in X$ and suppose $U = \prod_{\alpha\in J} U_\alpha$ is a basis element conatining $\mathbf{x}$. Then all but finitely many of open sets U_α equal X_α. Let $K \subset J$ be the finite set such that $U_k \neq X_k$ for $k \in K$ and $U_\alpha = X_\alpha$ for $\alpha \in J - K$. Then the point $\mathbf{b} = (b_\alpha)$, where $b_\alpha = x_\alpha$ for $\alpha \in K$ and $b_\alpha = a_\alpha$ otherwise is a member of Y which is in U. Thus $\mathbf{x}$ is a member of the closure of Y, so $\bar{Y} = X$. Since Y is connected, X is connected by Theorem 23.4. ■

Problem 8

Let $Y \subset X$; let X and Y be connected. Show that if A and B form a separation of $X - Y$, then $Y \cup A$ and $Y \cup B$ are connected.

Solution: It suffices to prove that $Y \cup A$ is connected; the proof for $Y \cup B$ is similar. Assume $C \cup D$ is a separation of $Y \cup A$. Without loss of generality, suppose $Y \subset C$ since Y is connected. Then since D is disjoint from $C, D \subset A$. Now, $X = Y \cup A \cup B = C \cup D \cup B$. Note that $C \cup B$ and D are disjoint. Since $C \cup D$ is a separation of $Y \cup A$, no limit points of C lie in D. Since $A \cup B$ is a separation of $X - Y$, no limit point of B lies in $A \supset D$, so $\overline{C \cup B} = \bar{C} \cup \bar{B} = C \cup B$, so $C \cup B$ is closed. Similarly, no limit points of D lie in $C \cup B$, so $\bar{D} = D$, so D is closed, so its complement $C \cup B$ is open. So $C \cup B$ is open and closed, a conteadiction to the fact that X is connected. Thus $Y \cup A$ is connected, and the proof is complete. ■

Problem 9

Let $f : S^1 \to \mathbb{R}$ be a continuous map. Show there exists a point $\mathbf{x}$ of S^1 such that $f(\mathbf{x}) = f(-\mathbf{x})$.

Solution: Define a map $g : S^1 \to \mathbb{R}$ by $g(\mathbf{x}) = f(\mathbf{x}) - f(-\mathbf{x})$. Let $\mathbf{x}_0 = (1, 0)$, and notice that $g(-\mathbf{x}_0) = -g(\mathbf{x}_0)$. If $g(\mathbf{x}_0) = 0$, then we are done. Otherwise, the equation $g(-\mathbf{x}_0) = -g(\mathbf{x}_0)$ tells us that $g(\mathbf{x}_0)$ and $g(-\mathbf{x}_0)$ have opposite signs. Thus by theorem 24.3, we conclude that there exists some point $\mathbf{y} \in S_1$ such that $g(\mathbf{y}) = 0$. Then for this $\mathbf{y}$, we have $f(\mathbf{y}) = f(-\mathbf{y})$, as desired. ■

Problem 10

Show that if U is an open connected subspace of $\mathbb{R}^2$, then U is path connected. **Hint**: Show that given $x_0 \in U$, the set of points that can be joined to x_0 by a path in U is both open and closed in U.

Solution: Let $x_0 \in U$ be arbitrary, and let $V \subset U$ be the set of points which can be joined to x_0 by a path in U. Let $x \in V$. Then since U is open, there exists some open ball $B(x, \varepsilon) \subset U$. Then, as we saw in the class, $B(x, \varepsilon)$ is path connected. Thus if we have $y \in B(x, \varepsilon)$, then there exists some path from x to y; joining this path with a path from x_0 to x yields a path from x_0 to y. Hence $y \in V$, and hence $B(x, \varepsilon) \subset V$. Since $x \in V$ was arbitrary, V is open. Now let $z \in U$ be a limit point of V, and let δ be such that $B(z, \delta) \subset U$. Then $B(z, \delta) \cap V \neq \varnothing$ since z is a limit point; let $z_0 \in B(z, \delta) \cap V$. Then since $z_0 \in V$, there exists a path from x_0 to z_0; since $z_0 \in B(z, \delta)$ and $B(z, \delta)$ is path connected, there exists a path from z_0 to z. Joining these two paths yields a path from x_0 to z, so that $z \in V$. Since z was arbitrary, we conclude that V contains all its limit points in U, so that V is closed in U. Since U is connected, the only sets that are both open and closed in it are $\varnothing$ and U, and $x_0 \in V$, so $V \neq \varnothing$. Therefore, we conclude that $V = U$. Thus x_0 can be connected via a path to every point of U; since x_0 was arbitrary, we conclude that U is path connected. ■

Problem 11

If A is a connected subspace of X, does it follow that $\operatorname{Int} A$ and $\operatorname{Bd} A$ are connected? Does the converse hold? Justify your answers.

Solution: If A is connected, the interior of A need not be. Consider $A = \left(\overline{\mathbb{R}}_+ \times \overline{\mathbb{R}}_+\right) \cup \left(\overline{\mathbb{R}}_- \times \overline{\mathbb{R}}_-\right)$ as a subspace of $\mathbb{R}^2$, where $\overline{\mathbb{R}}_-$ is the set of non-positive real numbers. Furthermore, if A is connected, the boundary of A need not be. Consider $A = [-1, 1]$ as a subspace of $\mathbb{R}$. The converse does not hold either. Let $X = \mathbb{R}^2$, and let

$$B = \left\{ x \times y \in \mathbb{R}^2 : x = 0 \text{ or } y = 0, \text{ but not both } \right\}.$$

Then B is the set of standard axes with the origin removed. Notice that $A = B \cup (\mathbb{R}_+ \times \mathbb{R}_+)$isn't connected. However, $\operatorname{int} A = \mathbb{R}_+ \times \mathbb{R}_+$ and $\operatorname{Bd} A = B \cup \{0\}$ are both connected.

If you want a simpler counterexample to the converse, and do not care if X is connected or not, let $X = [-2, -1] \cup [1, 2]$ and $A = [-2, -1] \cup \{1\}$. Notice that A is not connected, but $\operatorname{int} A = [-2, -1]$ and $\operatorname{Bd} A = \{1\}$ are both connected. ■